

Engineering Report - Q3 2024

Prepared by the Engineering Planning Office for the steering committee. This document summarizes the deliverables of the third quarter, the resource usage, and the outlook for Q4.

1. Delivered work packages

Nine of eleven planned work packages shipped in Q3. The two deferred packages (WP-04 telemetry gateway, WP-09 field calibration kit) were moved to Q4 with an approved scope change.

WP	Title	Effort (h)	Status
WP-01	Firmware 2.4 baseline	320	Done
WP-02	Power supply redesign	410	Done
WP-03	Sensor calibration rig	260	Done
WP-05	HMI software 3.1	380	Done
WP-06	EMC pre-compliance campaign	190	Done
WP-07	Packaging redesign	120	Done
WP-08	Field trial documentation	140	Done
WP-10	Spare parts portfolio	90	Done
WP-11	Service documentation	230	Done

2. Resource usage

Engineering capacity was 2,140 hours in Q3; 1,930 hours were booked against work packages, 120 hours went to support of the production ramp, and 90 hours were unassigned. Overtime stayed within the approved envelope of 5 %.

3. Risks and mitigations

The supply chain for the new power module remains the top risk. The second-source qualification is under way, the target date for full second-sourcing is 2025-02-01. The EMC campaign confirmed margins above 6 dB on all conducted and radiated tests.

The HMI software team reported a usability finding from the field trial regarding the alarm list; the fix is scheduled for the 3.1.2 maintenance release in October.

4. Outlook for Q4

Q4 focuses on the two deferred packages, the production ramp of the redesigned power supply, and the preparation of the 2025 platform review. A mid-quarter checkpoint is scheduled for calendar week 46, and the final steering committee

meeting of the year takes place on 2024-12-17.

The engineering planning office will publish the Q4 work package breakdown by 2024-10-08. Any remaining questions can be directed to the program office; the escalation path is unchanged from the Q2 report.

Approved by the engineering director on 2024-10-02, revision A. The next report in this series covers Q4 2024 and is due on 2025-01-10.